



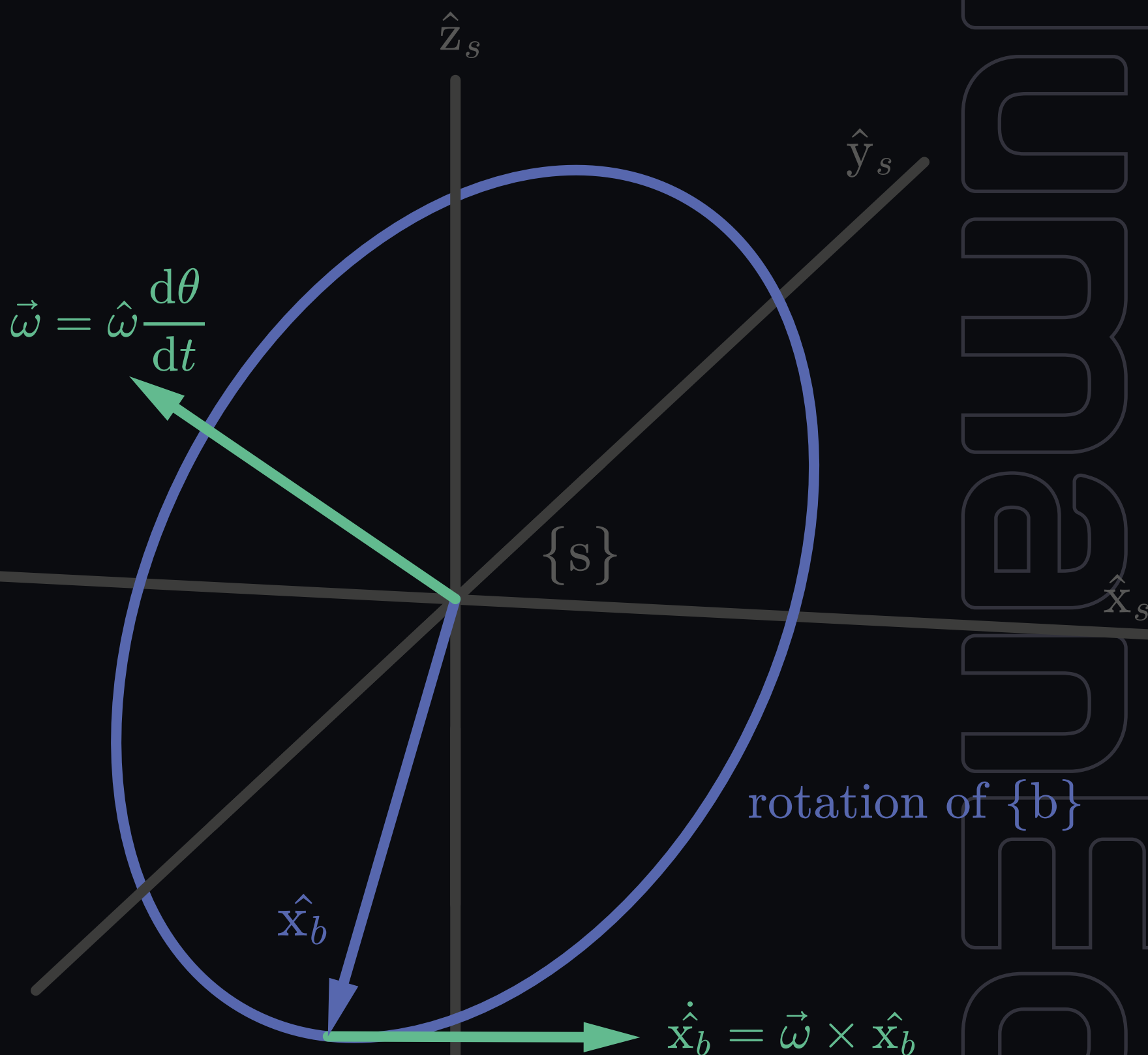
The angular velocity vector, $\vec{\omega} \in \mathbb{R}^3$, of a rotating body frame $\{b\}$ within a space frame $\{s\}$ can be described via its axis of rotation $\hat{\omega} \in \mathbb{R}^3$ ($\|\hat{\omega}\| = 1$) and the angle (displacement) of the rotation, $\theta \in \mathbb{R}$, such that:

$$\vec{\omega} = \hat{\omega} \dot{\theta} = \hat{\omega} \frac{d\theta}{dt}$$

Which for each axis of $\{b\}$ can be expressed as a set of equations:

$$\dot{\hat{x}}_b = \vec{\omega} \times \hat{x}_b, \quad \dot{\hat{y}}_b = \vec{\omega} \times \hat{y}_b, \quad \dot{\hat{z}}_b = \vec{\omega} \times \hat{z}_b$$

ANGULAR VELOCITY



Taking $R(t) \in SO(3)$ as the rotation matrix describing the orientation of the body frame at time t , it can be seen that:

$$\dot{\vec{r}}_i = \vec{\omega}_s \times \vec{r}_i$$

Where $\vec{r}_i$ ($i = 1, 2, 3$) are the columns of $R(t)$ and $\vec{\omega}_s \in \mathbb{R}^3$ is the angular velocity $\vec{\omega}$ expressed in the space frame coordinates. These three equations can be rearranged into a single 3×3 matrix, giving:

$$\dot{R} = [\vec{\omega}_s \times \vec{r}_1 \quad \vec{\omega}_s \times \vec{r}_2 \quad \vec{\omega}_s \times \vec{r}_3] = [\vec{\omega}_s] R$$

where $[\vec{\omega}_s]$ is the **skew-symmetric** matrix representation of $\vec{\omega}_s$.

The **skew-symmetric** matrix of a vector $\vec{x} = [x_1 \quad x_2 \quad x_3]^T \in \mathbb{R}^3$ is defined as:

$$[\vec{x}] = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$$

And holds the property that:

$$[\vec{x}] = -[\vec{x}]^T$$